EXPERIMENT-18:

Implement Perceptron based IRIS classification

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
# Upload Dataset
print("Upload Iris.csv")
uploaded = files.upload()
# Read Uploaded File
filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)
# Display Dataset
print("\nFirst 5 Rows:")
print(df.head())
print("\nDataset Shape:")
print(df.shape)
print("\nMissing Values:")
print(df.isnull().sum())

# Remove Id column (if present)
if 'Id' in df.columns:
    df = df.drop('Id', axis=1)
# Features and Target
X = df.drop('Species', axis=1)
y = df['Species']
# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
```

```
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
# Train Perceptron Model
model = Perceptron(
    max_iter=1000,
    random_state=42
)
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\n==============================")
print("Model Accuracy")
print("==============================")
print(f"{accuracy*100:.2f}%")
# Classification Report
print("\n==============================")
print("Classification Report")
print("==============================")
print(classification_report(y_test, y_pred))
# Confusion Matrix
print("\n==============================")
print("Confusion Matrix")
print("==============================")
```

print(confusion_matrix(y_test, y_pred))

Predict First Sample

sample = X.iloc[[0]]

sample = scaler.transform(sample)

prediction = model.predict(sample)

print("\n==============================")

print("Prediction for First Flower")

print("==============================")

print("Predicted Species:", prediction[0])

Output:

```
Upload Iris.csv
Choose Files Iris[1].csv
Iris[1].csv(text/csv) - 5107 bytes, last modified: 8/7/2026 - 100% done
Saving Iris[1].csv to Iris[1].csv

First 5 Rows:
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0   1            5.1           3.5            1.4           0.2  Iris-setosa
1   2            4.9           3.0            1.4           0.2  Iris-setosa
2   3            4.7           3.2            1.3           0.2  Iris-setosa
3   4            4.6           3.1            1.5           0.2  Iris-setosa
4   5            5.0           3.6            1.4           0.2  Iris-setosa

Dataset Shape:
(150, 6)

Missing Values:
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64

==============================
Model Accuracy
==============================
86.67%

==============================
Classification Report
==============================
                 precision    recall  f1-score   support

    Iris-setosa       0.83      1.00      0.91        10
Iris-versicolor       0.88      0.70      0.78        10
 Iris-virginica       0.90      0.90      0.90        10

       accuracy                           0.87        30
      macro avg       0.87      0.87      0.86        30
   weighted avg       0.87      0.87      0.86        30


==============================
Confusion Matrix
==============================
[[10  0  0]
 [ 2  7  1]
 [ 0  1  9]]

==============================
Prediction for First Flower
==============================
Predicted Species: Iris-setosa
```